

Network to Link Layer Transition

Data vs. Control Plane & Introduction to the Link Layer

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 - The Data Plane: Forwarding
 - The Control Plane: Routing & SDN
- **Part 2: Transitioning Down the Stack**
 - End-to-end delivery vs. Node-to-node delivery.
- **Part 3: The Link Layer I (Ch. 6)**
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 - Link Layer Services and Implementation
 - Error Detection and Correction (Parity & CRC)
 - Multiple Access Links and Protocols (MAC)

The Network Layer: Two Interacting Planes

To fully understand how the Internet moves packets, we divide the Network Layer into two interacting planes:

The Data Plane (Chapter 4)

The local, **per-router** function. It determines how a datagram arriving on an input port is forwarded to the correct output port. Implemented entirely in hardware (ASICs) for nanosecond speed.

The Control Plane (Chapter 5)

The **network-wide** logic. It determines how datagrams are routed end-to-end along paths from source to destination. Implemented in software operating at millisecond/second timescales.

Transitioning Down the Stack

- **Where we are:** The Network Layer (Control Plane) has calculated the best path from Host A to Host B across the globe. The Data Plane moves the packet through each router's hardware.
- **The Missing Piece:** The Network Layer views a path as a series of routers. But how does a packet physically travel over the specific cable or wireless frequency connecting Router 1 to Router 2?
- **The Answer:** The Link Layer. Its basic service is to move a datagram from one node to an adjacent node over a single communication link.

Terminology and the Travel Agent Analogy

- **Terminology:**

- Devices (hosts, routers, switches, WiFi access points) are **nodes**.
- Communication channels connecting adjacent nodes are **links**.
- Link-layer packets are **frames**.

- **The Analogy:** Consider a tourist traveling from Princeton to Lausanne.

- Tourist = Datagram.
- Travel Agent = Routing Protocol (Network Layer).
- Transportation Segments (Limo, Plane, Train) = Links.
- Transportation Mode (The companies running the limo/plane) = Link-Layer Protocol.

What does the Link Layer actually do?

- 1 **Framing:** Encapsulates the network-layer datagram within a link-layer frame, consisting of a data field and header fields.
- 2 **Link Access (MAC):** Specifies the rules by which a frame is transmitted onto the link. Critical for coordinating frame transmissions on shared broadcast links.
- 3 **Reliable Delivery:** Guarantees moving each datagram across the link without error. Often used for high-error wireless links, but considered unnecessary overhead for low-error wired links (fiber, coax).
- 4 **Error Detection & Correction:** Detects bit errors caused by signal attenuation and noise. Usually implemented in hardware.

Where is the Link Layer Implemented?

- **The Network Adapter:** For the most part, the link layer is implemented on a chip called the network adapter, or network interface controller (NIC).
- **Hardware Centric:** It implements framing, link access, and error detection in dedicated hardware.
- **Software Component:** Part of the link layer is implemented in software running on the host's CPU to handle interrupts and pass datagrams up to the network layer.
- **The Boundary:** The link layer is the specific place in the protocol stack where software meets hardware.

A Closer Look: Error Detection

Because physical signals degrade, the Link Layer must catch flipped bits.

- **Single Parity Checks:** Uses one additional bit to ensure an even or odd number of 1s. Fails heavily during "burst" errors.
- **Two-Dimensional Parity:** Arranges data in a matrix. Can detect and actually *correct* single bit errors. This is known as Forward Error Correction (FEC).
- **Internet Checksum:** Treats data as 16-bit integers and sums them. Used in Transport/Network layers because it is fast for software, but provides weak protection compared to hardware link-layer methods.

Cyclic Redundancy Check (CRC)

- CRC is the error-detection standard used widely in Ethernet and Wi-Fi.
- Also known as polynomial codes, it uses modulo-2 arithmetic (bitwise XOR without carries).
- **The Logic:** The sender and receiver agree on a generator bit pattern, G . The sender appends r bits, R , to the data D so that the resulting pattern is exactly divisible by G .
- **The Formula:** The sender calculates R such that $D \cdot 2^r \text{ XOR } R = nG$.
- **The Check:** The receiver divides the received bits by G . If the remainder is non-zero, an error has occurred.

Multiple Access Links: The Cocktail Party

There are two types of network links: point-to-point (single sender/receiver) and broadcast (multiple nodes on a shared channel).

The "Cocktail Party" Problem

A broadcast channel is like a cocktail party. If multiple people (nodes) speak (transmit) at the exact same time, their signals become inextricably tangled together, resulting in a **collision** where all frames are lost.

To solve this, we use **Multiple Access Protocols** to coordinate access. There are three main categories:

- 1 Channel Partitioning (TDM, FDM, CDMA)
- 2 Random Access (ALOHA, CSMA/CD)
- 3 Taking-Turns (Polling, Token passing)

Random Access: CSMA/CD (Ethernet)

Unlike TDM/FDM, Random Access protocols always transmit at the full channel rate.

- **Carrier Sense (Listen before speaking):** A node listens to the channel. If it is busy, it waits until the channel is idle.
- **Collision Detection (Stop if interrupted):** A node listens while transmitting. If it detects interference, it immediately aborts transmission to avoid wasting time.
- **Random Backoff:** After a collision, nodes wait a random delay before retransmitting. Ethernet uses **binary exponential backoff**, exponentially increasing the maximum possible delay interval with each successive collision.